

DEPARTMENT OF STATISTICAL SCIENCE
2026 - ADVANCED REGRESSION TOPICS
TUTORIAL EXERCISES 2

1. (a) Derive $\hat{\beta}^{\text{ridge}}$, the coefficient estimates for a ridge regression model. Start by writing down the regression model, describing each of its elements including their dimensions.
(b) Show that $\hat{\beta}^{\text{ridge}}$ is a biased estimator when $\lambda > 0$.
(c) Find $\text{Var} [\hat{\beta}^{\text{ridge}}]$, the variance of the ridge regression estimator.
(d) In your own words, describe how values of λ affect the bias and the variance of your estimator in part (a).
2. Consider the linear regression model

$$\mathbf{y} = \mathbf{X}\beta + \varepsilon$$

- (a) Show that minimizing the residual sum of squares RSS is equivalent to finding the residual vector that is orthogonal to the column space of $\mathbf{X}$.
 - (b) Also show how $MSE(\hat{\beta}^{\text{ls}})$ can be decomposed into its two components.
 - (c) How does the ridge regression estimator compare to the ordinary least squares estimator in the following cases?
 - i. $\lambda \rightarrow 0$.
 - ii. The columns of the input matrix are orthonormal.
 - (d) In ridge regression estimation, what happens when $\lambda \rightarrow \infty$?
3. (a) Consider $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3$, a non-orthogonal basis for a given vector space. Use the Gram-Schmidt procedure to find an orthogonal basis $\mathbf{z}_1, \mathbf{z}_2, \mathbf{z}_3$ for the given vector space. Make sure to sketch a geometric representation to help visualize the Gram-Schmidt steps.
(b) Use your answer in part (a) to show that the matrix $\mathbf{X}$ with columns $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3$, can be written as

$$\mathbf{X} = \mathbf{Q}\mathbf{R}$$

where $\mathbf{R}$ is an upper triangular matrix, and the columns of the matrix $\mathbf{Q}$ are orthonormal.

- (c) Now consider the linear regression model

$$\mathbf{y} = \mathbf{X}\beta + \varepsilon$$

where $\mathbf{y}$ is an N -dimensional vector, $\mathbf{X}$ is an $N \times 3$ design matrix, and ε is an N -dimensional random vector with mean $\mathbf{0}$ and variance-covariance matrix $\sigma^2 \mathbf{I}$. the vector $\mathbf{y}$ as well as the columns of $\mathbf{X}$ have been standardized. Find the least squares estimator $\hat{\beta}^{\text{ls}}$ and the predicted response $\hat{\mathbf{y}}$ in terms of $\mathbf{Q}$ and $\mathbf{R}$.

4.
 - (a) As regression shrinkage methods, what is the main difference between Lasso and Ridge regression? Elaborate.
 - (b) Give in Lagrangian form, the expression of the Lasso estimator $\hat{\beta}^{\text{lasso}}$
 - (c) Provide one advantage and one disadvantage of the lasso.